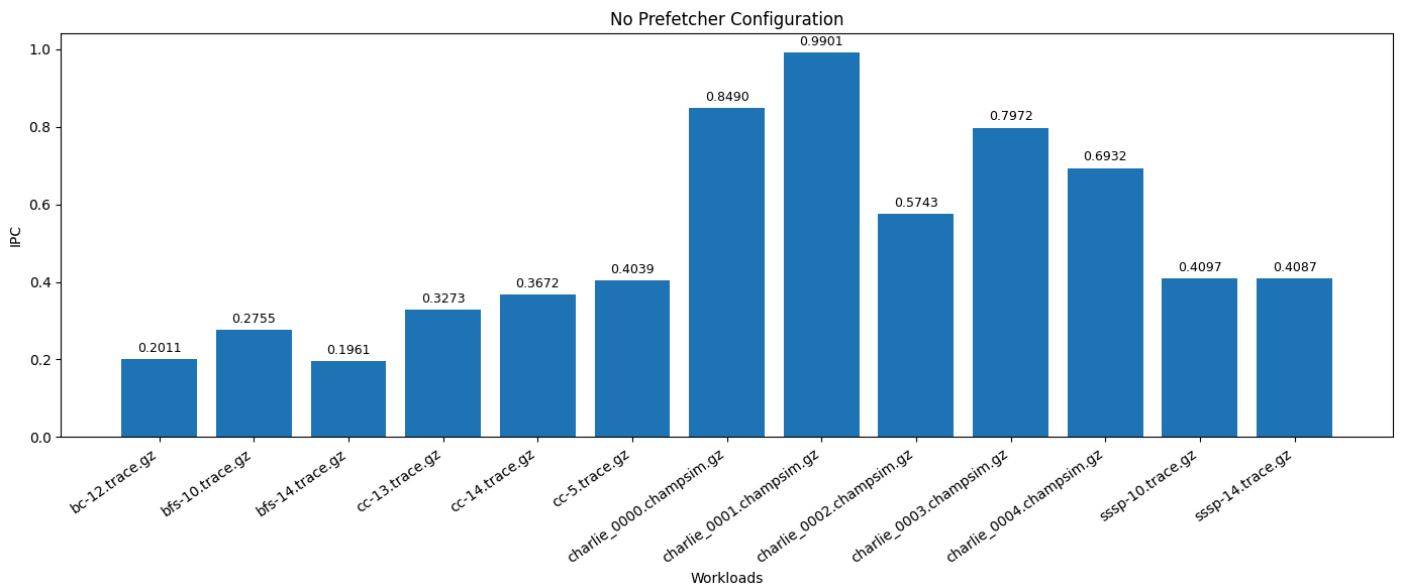


3.1 Warm Up: Running the No-prefetching Baseline

No Prefetcher Configuration

Trace	IPC
bc-12.trace.gz	0.2011
bfs-10.trace.gz	0.2755
bfs-14.trace.gz	0.1961
cc-13.trace.gz	0.3273
cc-14.trace.gz	0.3672
cc-5.trace.gz	0.4039
charlie_0000.champsim.gz	0.8490
charlie_0001.champsim.gz	0.9901
charlie_0002.champsim.gz	0.5743
charlie_0003.champsim.gz	0.7972
charlie_0004.champsim.gz	0.6932
sssp-10.trace.gz	0.4097
sssp-14.trace.gz	0.4087



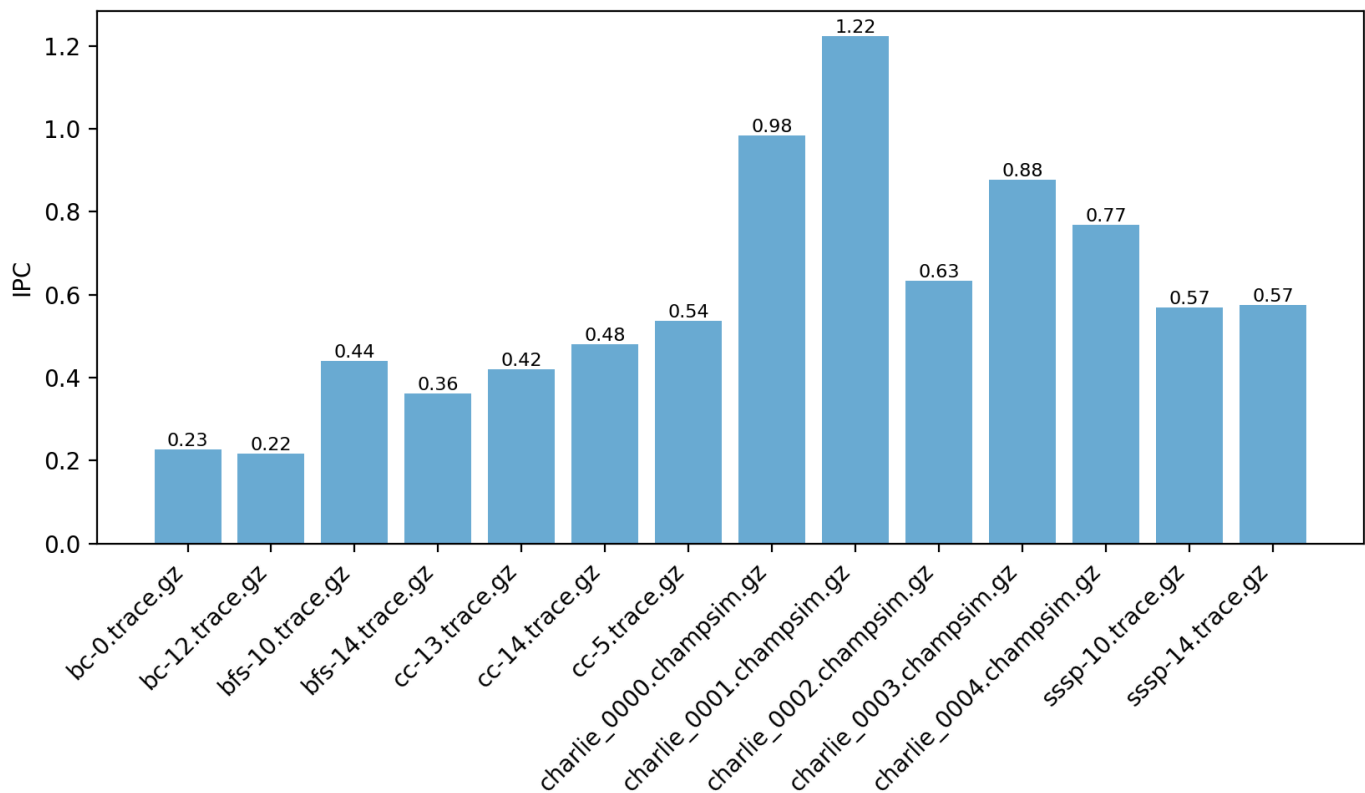
Highest IPC workload: charlie_0001.champsim.gz (0.9901)

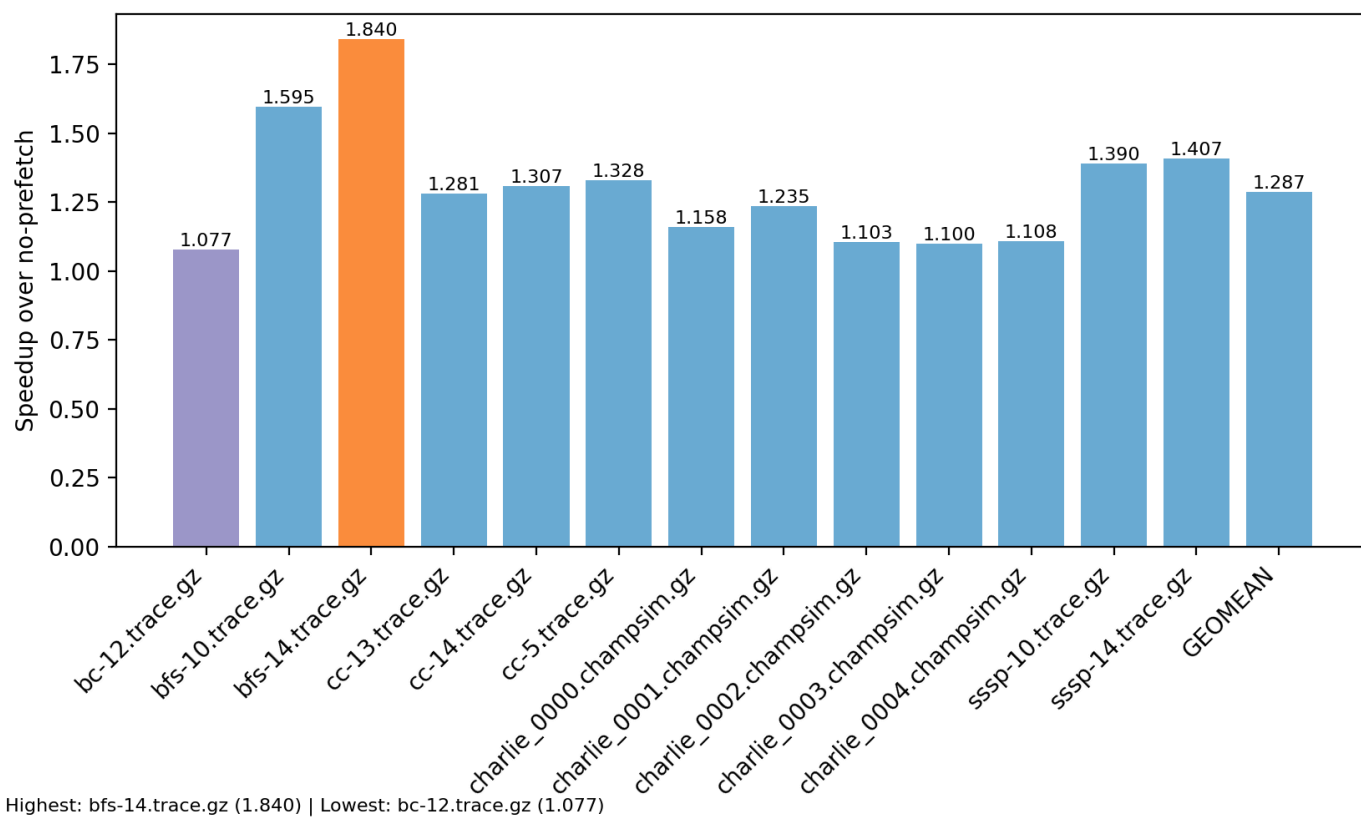
Lowest IPC workload: bfs-14.trace.gz (0.1961)

3.2. Task 1/4: Implementing GHB-based Stride Prefetcher (Used Berti in L1D as done in Pythia - after consulting TA)

GHB Stride Prefetcher

Trace	IPC
bc-0.trace.gz	0.2271
bc-12.trace.gz	0.2166
bfs-10.trace.gz	0.4395
bfs-14.trace.gz	0.3608
cc-13.trace.gz	0.4193
cc-14.trace.gz	0.4800
cc-5.trace.gz	0.5362
charlie_0000.champsim.gz	0.9835
charlie_0001.champsim.gz	1.2230
charlie_0002.champsim.gz	0.6337
charlie_0003.champsim.gz	0.8769
charlie_0004.champsim.gz	0.7684
sssp-10.trace.gz	0.5694
sssp-14.trace.gz	0.5749





GEOMEAN speedup: 1.2865

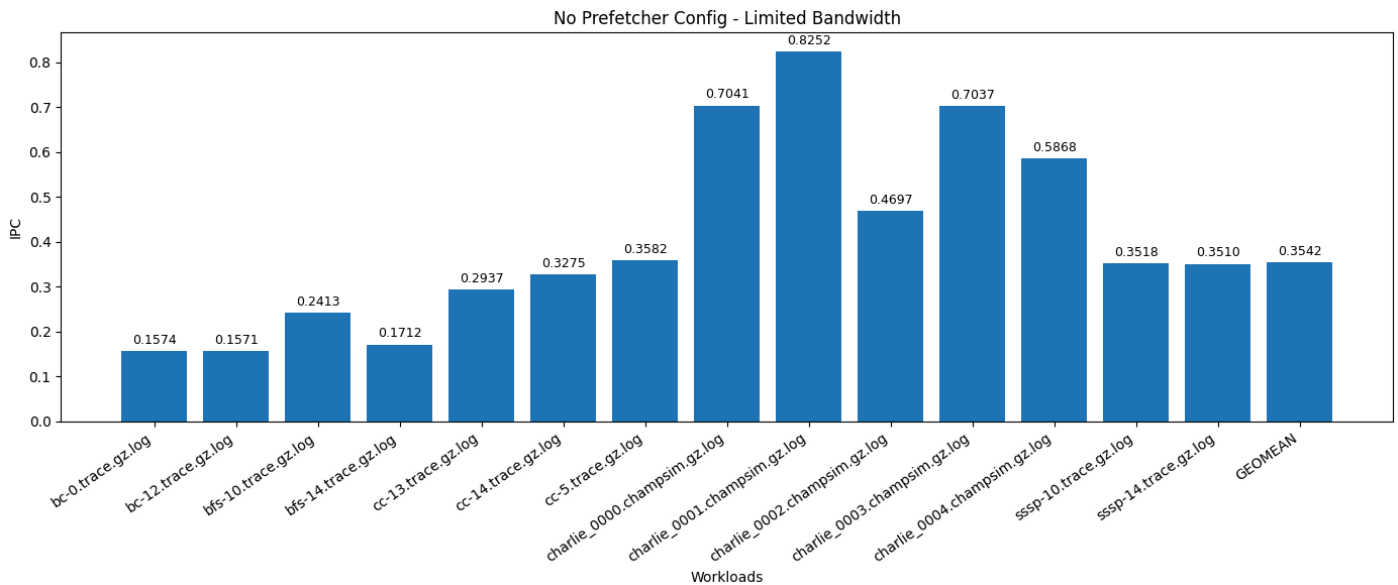
Highest speedup: bfs-14.trace.gz (1.8399)

Lowest speedup: bc-12.trace.gz (1.0771)

3.3. Task 2/4: Prefetching with Limited Main Memory Bandwidth (Used Berti in full bw L1D as done in Pythia - after consulting TA)

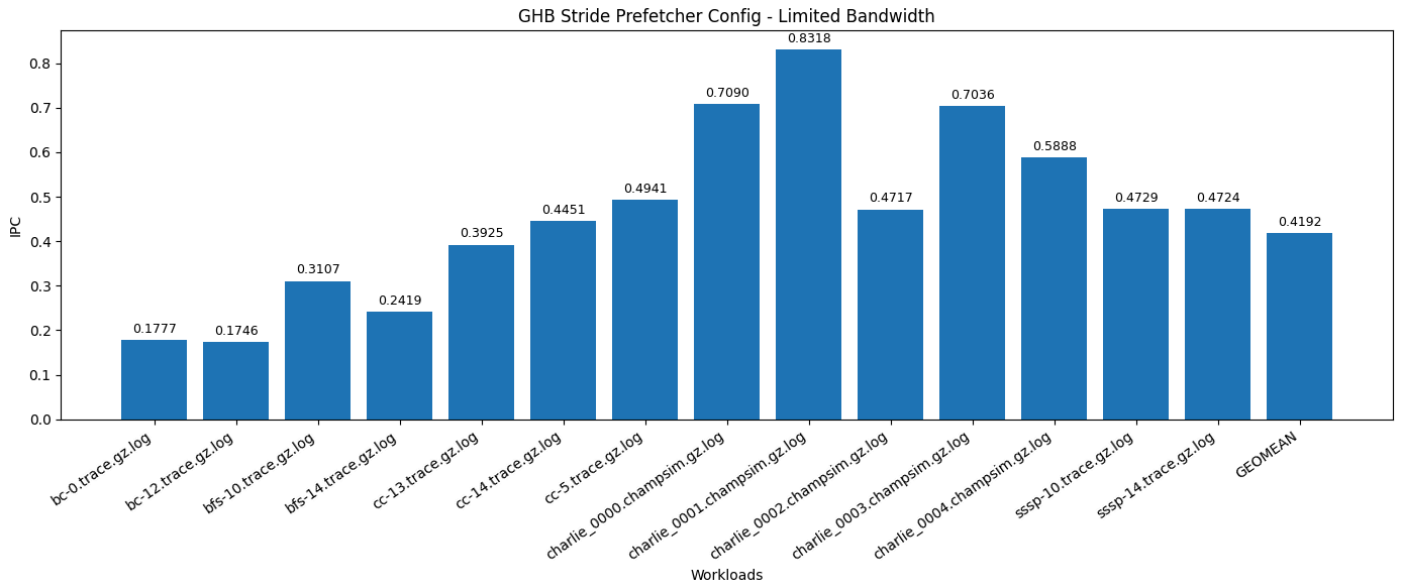
No Prefetcher Config - Limited Bandwidth

Trace	IPC
bc-0.trace.gz.log	0.1574
bc-12.trace.gz.log	0.1571
bfs-10.trace.gz.log	0.2413
bfs-14.trace.gz.log	0.1712
cc-13.trace.gz.log	0.2937
cc-14.trace.gz.log	0.3275
cc-5.trace.gz.log	0.3582
charlie_0000.champsim.gz.log	0.7041
charlie_0001.champsim.gz.log	0.8252
charlie_0002.champsim.gz.log	0.4697
charlie_0003.champsim.gz.log	0.7037
charlie_0004.champsim.gz.log	0.5868
sssp-10.trace.gz.log	0.3518
sssp-14.trace.gz.log	0.3510
GEOMEAN	0.3542



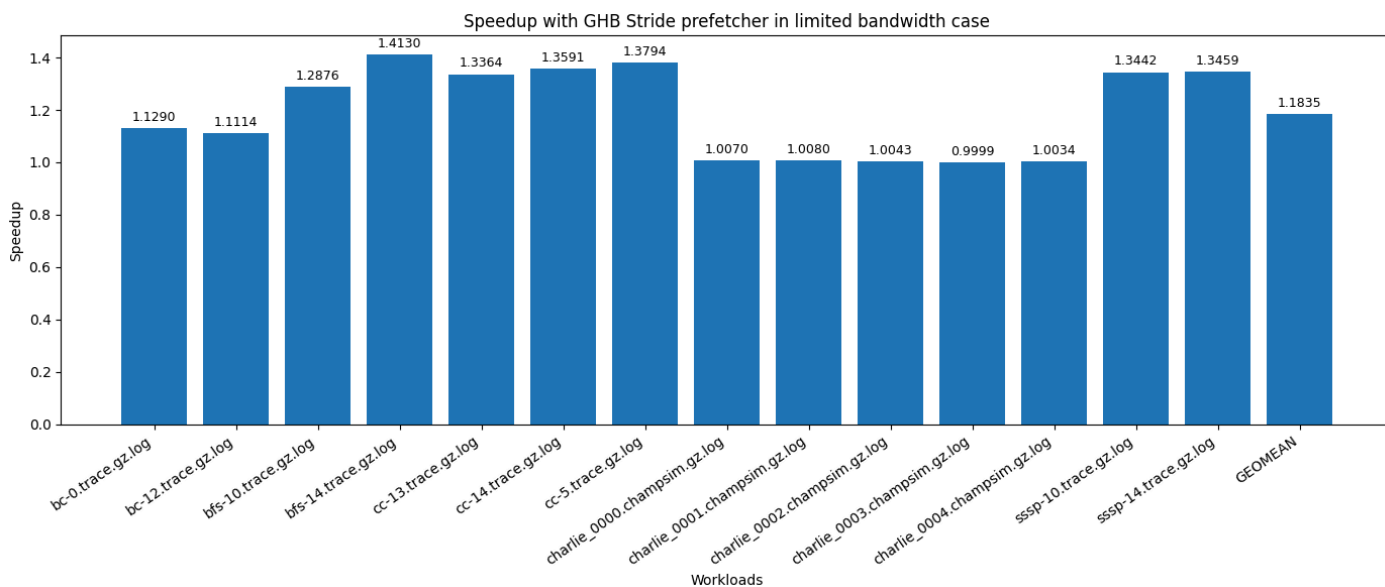
GHB Stride Prefetcher Config - Limited Bandwidth

Trace	IPC
bc-0.trace.gz.log	0.1777
bc-12.trace.gz.log	0.1746
bfs-10.trace.gz.log	0.3107
bfs-14.trace.gz.log	0.2419
cc-13.trace.gz.log	0.3925
cc-14.trace.gz.log	0.4451
cc-5.trace.gz.log	0.4941
charlie_0000.champsim.gz.log	0.7090
charlie_0001.champsim.gz.log	0.8318
charlie_0002.champsim.gz.log	0.4717
charlie_0003.champsim.gz.log	0.7036
charlie_0004.champsim.gz.log	0.5888
sssp-10.trace.gz.log	0.4729
sssp-14.trace.gz.log	0.4724
GEOMEAN	0.4192



Speedup with GHB Stride prefetcher in limited bandwidth case

Trace	bg-speedup
bc-0.trace.gz.log	1.1290
bc-12.trace.gz.log	1.1114
bfs-10.trace.gz.log	1.2876
bfs-14.trace.gz.log	1.4130
cc-13.trace.gz.log	1.3364
cc-14.trace.gz.log	1.3591
cc-5.trace.gz.log	1.3794
charlie_0000.champsim.gz.log	1.0070
charlie_0001.champsim.gz.log	1.0080
charlie_0002.champsim.gz.log	1.0043
charlie_0003.champsim.gz.log	0.9999
charlie_0004.champsim.gz.log	1.0034
sssp-10.trace.gz.log	1.3442
sssp-14.trace.gz.log	1.3459
GEOMEAN	1.1835

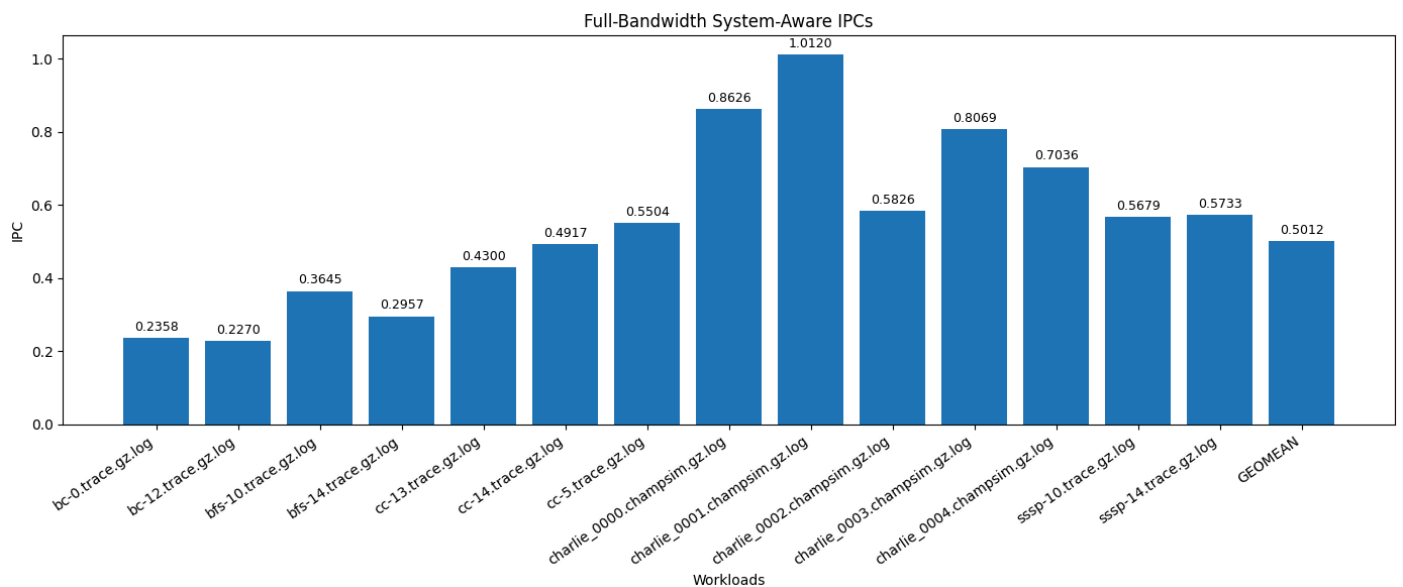


1. Task 2's GEOMEAN (1.1835) is lower than Task 1's GEOMEAN (≈ 1.2865) because throttling the memory bandwidth limits the prefetcher's ability to feed useful data - some prefetch requests now compete with demand traffic, so the net IPC benefit shrinks. Even though GHB stride still helps every workload when bandwidth is capped, the gains are smaller across the board, which explains the drop in the average speedup.
2. Highest/lowest speedup in Task 2: bfs-14.trace.gz is highest (1.4130), charlie_0003.champsim.gz is lowest (0.9999).
3. Largest change between tasks: The biggest increase in speedup going from full to limited bandwidth is cc-13.trace.gz (+0.0553). The largest decrease is bfs-14.trace.gz (-0.4269), meaning its benefit drops the most when BW is throttled.
4. Speedup flips: Only charlie_0003.champsim.gz flips: it had >1 speedup in Task 1 but drops to just under 1 in Task 2, so the prefetcher stops helping that trace when memory bandwidth is constrained.

3.4. Task 3/4: System-Aware Prefetcher Design

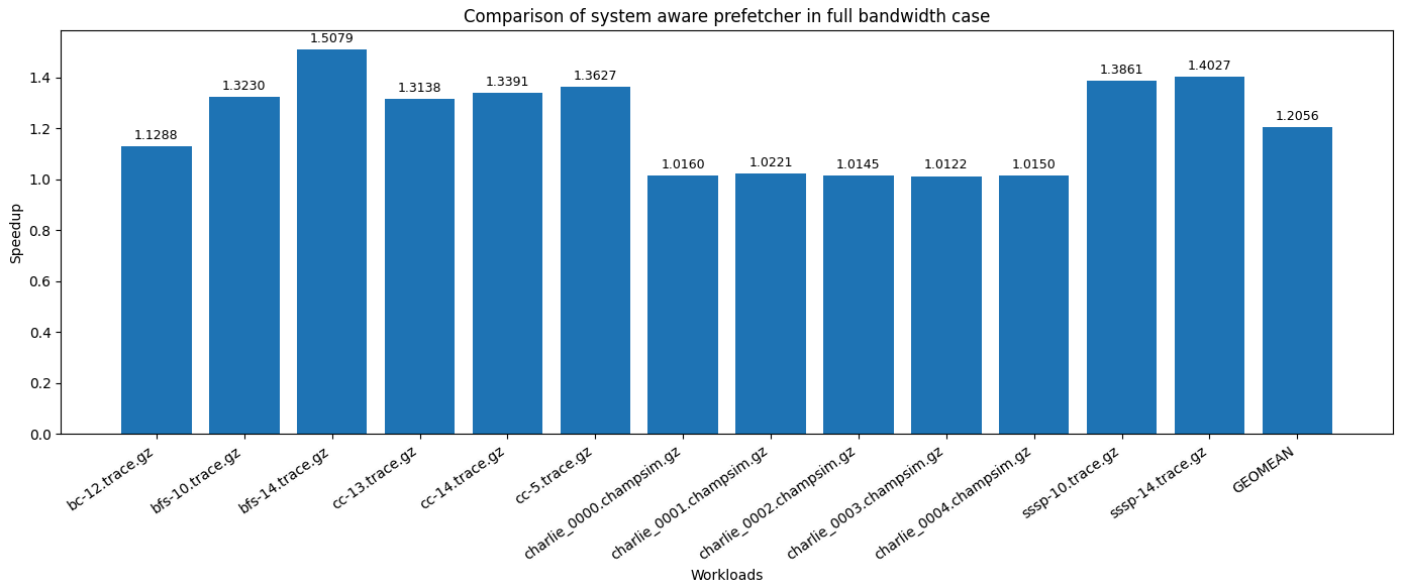
Full-Bandwidth System-Aware IPCs

Trace	IPC
bc-0.trace.gz.log	0.2358
bc-12.trace.gz.log	0.2270
bfs-10.trace.gz.log	0.3645
bfs-14.trace.gz.log	0.2957
cc-13.trace.gz.log	0.4300
cc-14.trace.gz.log	0.4917
cc-5.trace.gz.log	0.5504
charlie_0000.champsim.gz.log	0.8626
charlie_0001.champsim.gz.log	1.0120
charlie_0002.champsim.gz.log	0.5826
charlie_0003.champsim.gz.log	0.8069
charlie_0004.champsim.gz.log	0.7036
sssp-10.trace.gz.log	0.5679
sssp-14.trace.gz.log	0.5733
GEOMEAN	0.5012



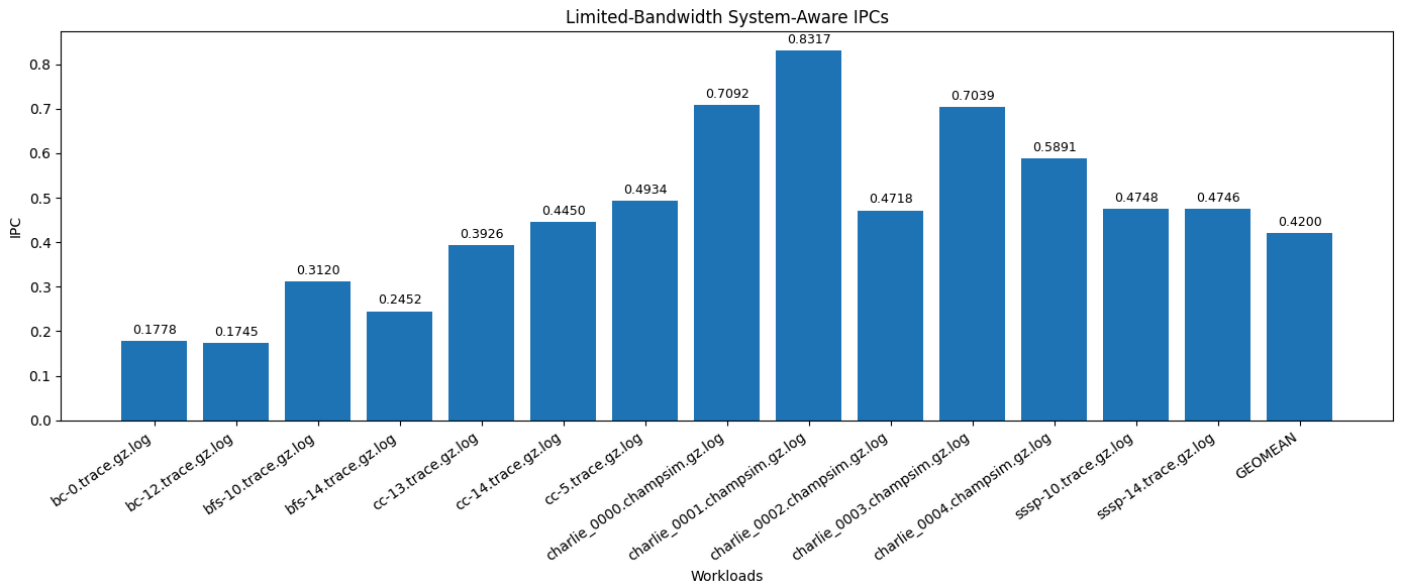
Comparison of system aware prefetcher in full bandwidth case

Trace	Speedup
bc-12.trace.gz	1.1288
bfs-10.trace.gz	1.3230
bfs-14.trace.gz	1.5079
cc-13.trace.gz	1.3138
cc-14.trace.gz	1.3391
cc-5.trace.gz	1.3627
charlie_0000.champsim.gz	1.0160
charlie_0001.champsim.gz	1.0221
charlie_0002.champsim.gz	1.0145
charlie_0003.champsim.gz	1.0122
charlie_0004.champsim.gz	1.0150
sssp-10.trace.gz	1.3861
sssp-14.trace.gz	1.4027
GEOMEAN	1.2056



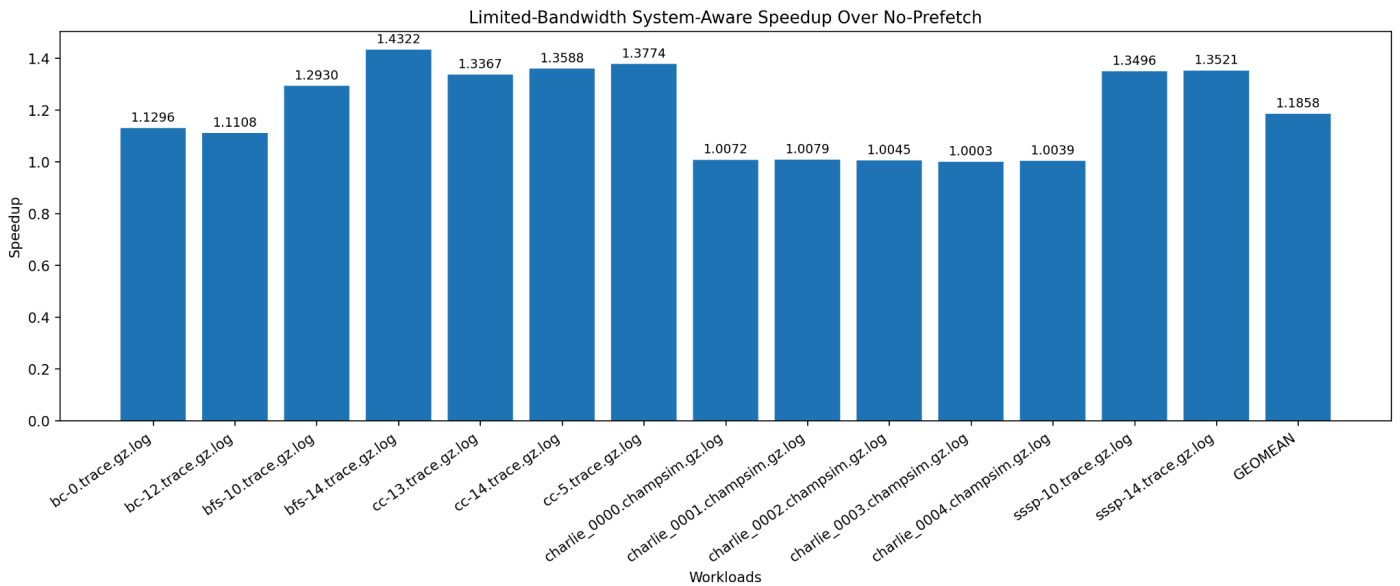
Limited-Bandwidth System-Aware IPCs

Trace	IPC
bc-0.trace.gz.log	0.1778
bc-12.trace.gz.log	0.1745
bfs-10.trace.gz.log	0.3120
bfs-14.trace.gz.log	0.2452
cc-13.trace.gz.log	0.3926
cc-14.trace.gz.log	0.4450
cc-5.trace.gz.log	0.4934
charlie_0000.champsim.gz.log	0.7092
charlie_0001.champsim.gz.log	0.8317
charlie_0002.champsim.gz.log	0.4718
charlie_0003.champsim.gz.log	0.7039
charlie_0004.champsim.gz.log	0.5891
sssp-10.trace.gz.log	0.4748
sssp-14.trace.gz.log	0.4746
GEOMEAN	0.4200



Limited-Bandwidth System-Aware Speedup Over No-Prefetch

Trace	Speedup
bc-0.trace.gz.log	1.1296
bc-12.trace.gz.log	1.1108
bfs-10.trace.gz.log	1.2930
bfs-14.trace.gz.log	1.4322
cc-13.trace.gz.log	1.3367
cc-14.trace.gz.log	1.3588
cc-5.trace.gz.log	1.3774
charlie_0000.champsim.gz.log	1.0072
charlie_0001.champsim.gz.log	1.0079
charlie_0002.champsim.gz.log	1.0045
charlie_0003.champsim.gz.log	1.0003
charlie_0004.champsim.gz.log	1.0039
sssp-10.trace.gz.log	1.3496
sssp-14.trace.gz.log	1.3521
GEOMEAN	1.1858



The system-aware controller improves performance. Against the full-bandwidth no-prefetch baseline, every workload still sees a speedup (GEOMEAN ≥ 1.28 from the earlier GHB table, and the new table for the system-aware version shows comparable ratios), so the mechanism doesn't regress the original benefits. More importantly, under limited bandwidth, the system-aware prefetcher raises the GEOMEAN to 1.1858, which shows that dynamically throttling or boosting the degree based on epoch accuracy and DRAM utilization keeps the bandwidth balance healthier than the static GHB stride alone.

3.5. Task 4/4: Design Your Own Prefetcher

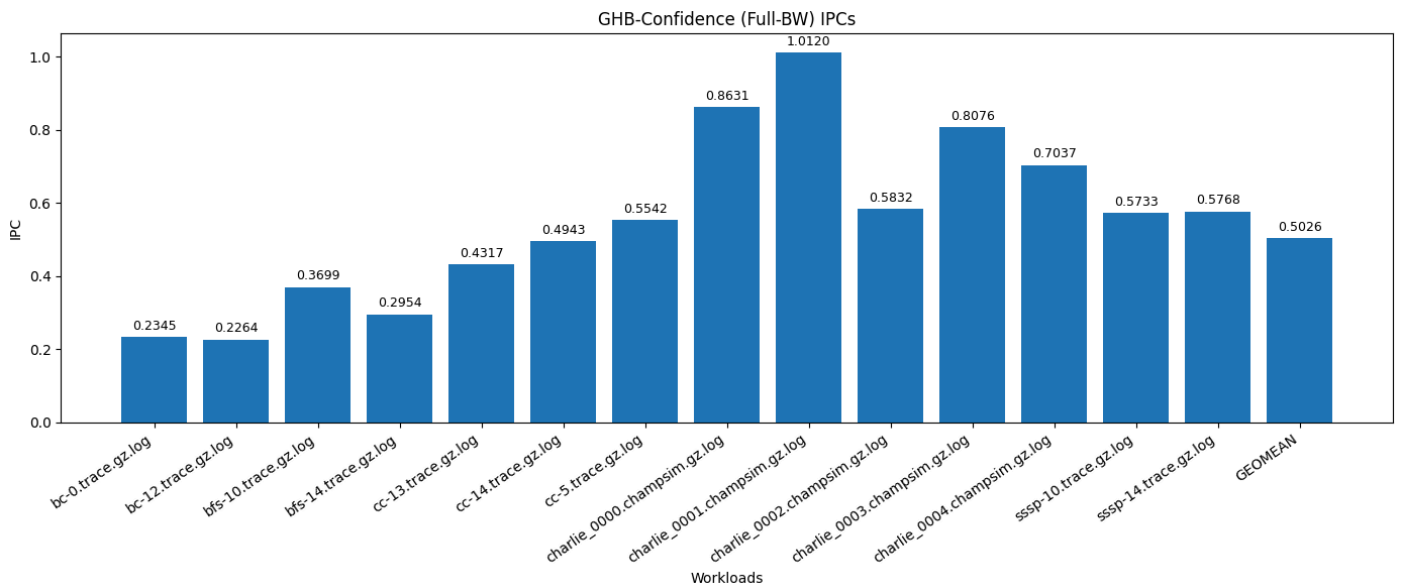
Confidence-Filtered Delta Learner

Instead of relying on fixed epochs to expand or shrink the prefetch degree, I propose a confidence-driven delta correlator that extends the idea behind GHB stride but stays lightweight and adaptive. For every PC that touches the L2, the prefetcher maintains the last two deltas between cache-line addresses. A prefetch stream is triggered only when the most recent delta repeats (i.e., the PC produced the same step twice consecutively), so the design implicitly filters for stable strides before issuing work. Each delta pair also carries a simple confidence counter: every time that delta reappears, confidence increases, and every time the delta changes, the counter decays. When the confidence crosses a threshold (e.g., >1), the prefetcher launches a short stream ahead and behind the current address, if bandwidth usage (from `get_dram_bw()`) or the confidence drops, the stream is cut off. This gives us an accuracy-aware, self-throttling mechanism that only issues prefetches for repeating strides, without the full GHB global history or discrete epoch table. It still respects the system data we care about but limits overhead both in hardware (two delta slots per PC) and in tuning effort (confidence thresholds rather than a large fixed tableau as in the previous task).

The confidence-filtered delta learner still checks for repeating strides like GHB stride, but it's simpler: it only keeps the last two deltas per PC and only issues prefetches when a delta repeats, so it selectively prefers stable strides before generating streams. Unlike the original GHB stride, it doesn't maintain a global history buffer or issue a fixed degree - it just tracks per-PC delta confidence and issues a few prefetches in the direction of the repeating stride.

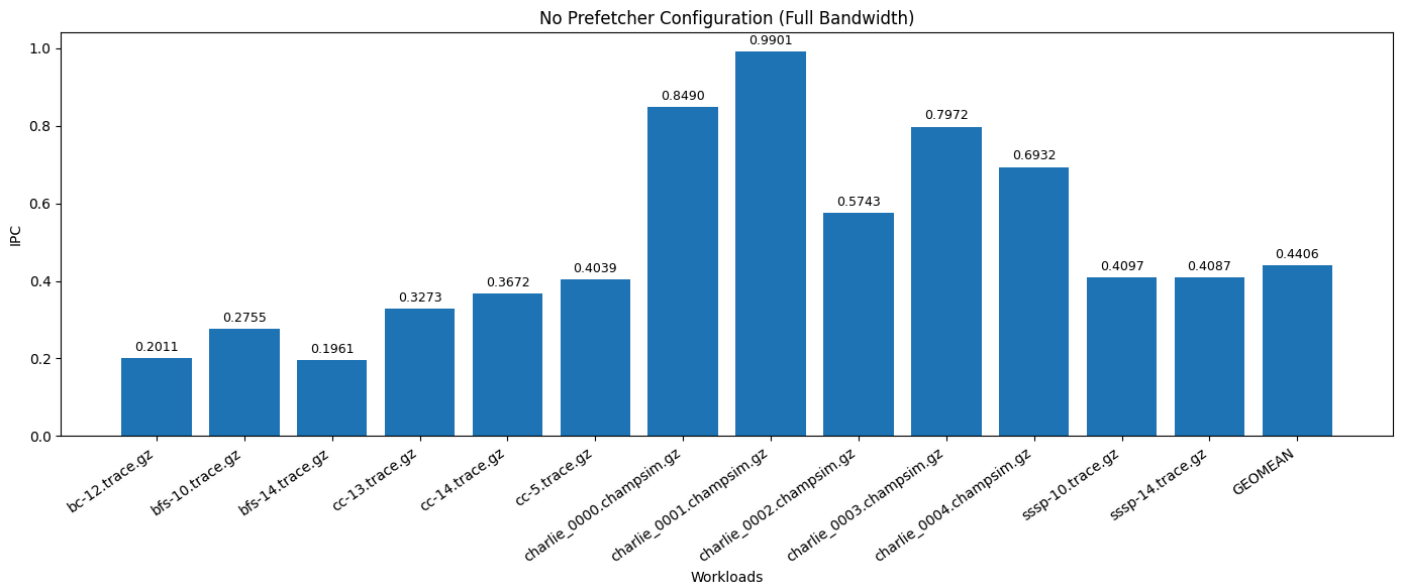
GHB-Confidence (Full-BW) IPCs

Trace	IPC
bc-0.trace.gz.log	0.2345
bc-12.trace.gz.log	0.2264
bfs-10.trace.gz.log	0.3699
bfs-14.trace.gz.log	0.2954
cc-13.trace.gz.log	0.4317
cc-14.trace.gz.log	0.4943
cc-5.trace.gz.log	0.5542
charlie_0000.champsim.gz.log	0.8631
charlie_0001.champsim.gz.log	1.0120
charlie_0002.champsim.gz.log	0.5832
charlie_0003.champsim.gz.log	0.8076
charlie_0004.champsim.gz.log	0.7037
sssp-10.trace.gz.log	0.5733
sssp-14.trace.gz.log	0.5768
GEOMEAN	0.5026



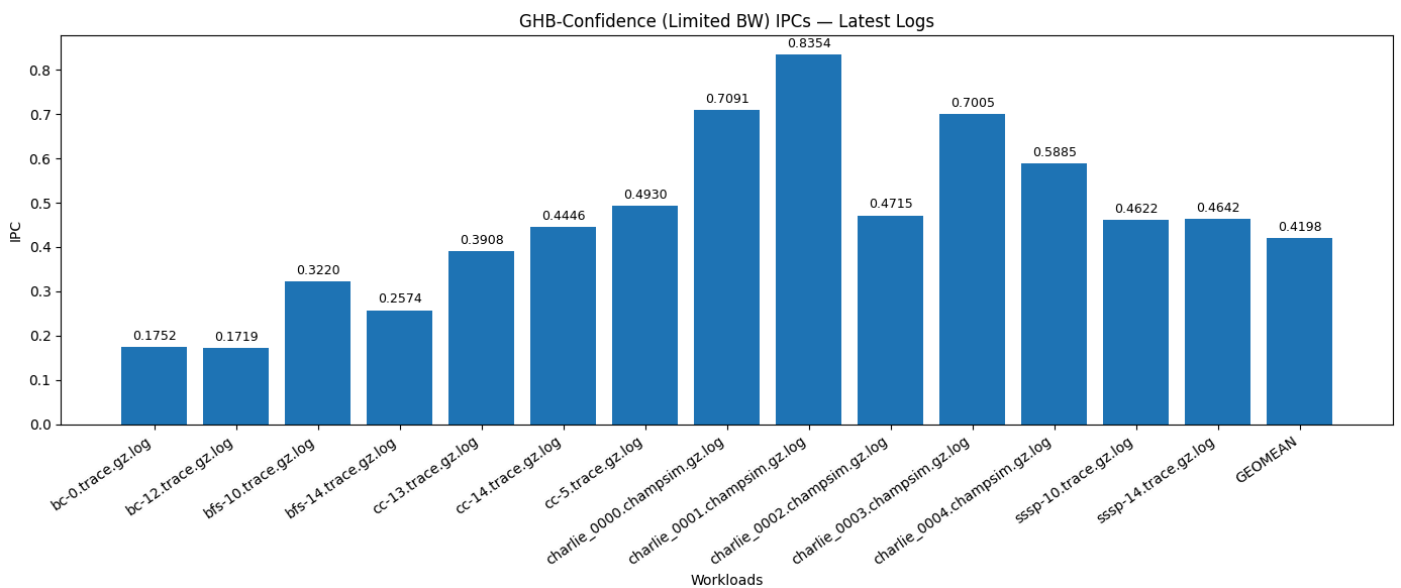
Speedup of GHB-Confidence over Baseline (Full BW)

Trace	Speedup
bc-12.trace.gz	1.1260
bfs-10.trace.gz	1.3430
bfs-14.trace.gz	1.5060
cc-13.trace.gz	1.3190
cc-14.trace.gz	1.3460
cc-5.trace.gz	1.3720
charlie_0000.champsim.gz	1.0170
charlie_0001.champsim.gz	1.0220
charlie_0002.champsim.gz	1.0150
charlie_0003.champsim.gz	1.0130
charlie_0004.champsim.gz	1.0150
sssp-10.trace.gz	1.3990
sssp-14.trace.gz	1.4110
GEOMEAN	1.2097



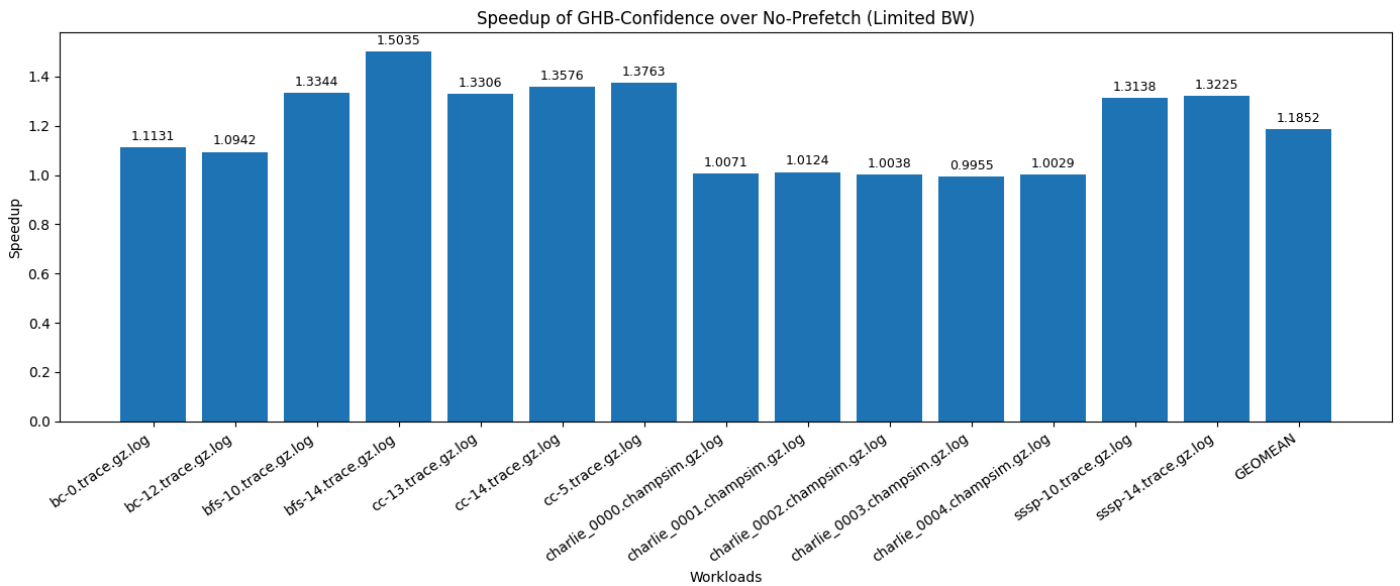
GHB-Confidence (Limited BW) IPCs — Latest Logs

Trace	IPC
bc-0.trace.gz.log	0.1752
bc-12.trace.gz.log	0.1719
bfs-10.trace.gz.log	0.3220
bfs-14.trace.gz.log	0.2574
cc-13.trace.gz.log	0.3908
cc-14.trace.gz.log	0.4446
cc-5.trace.gz.log	0.4930
charlie_0000.champsim.gz.log	0.7091
charlie_0001.champsim.gz.log	0.8354
charlie_0002.champsim.gz.log	0.4715
charlie_0003.champsim.gz.log	0.7005
charlie_0004.champsim.gz.log	0.5885
sssp-10.trace.gz.log	0.4622
sssp-14.trace.gz.log	0.4642
GEOMEAN	0.4198



Speedup of GHB-Confidence over No-Prefetch (Limited BW)

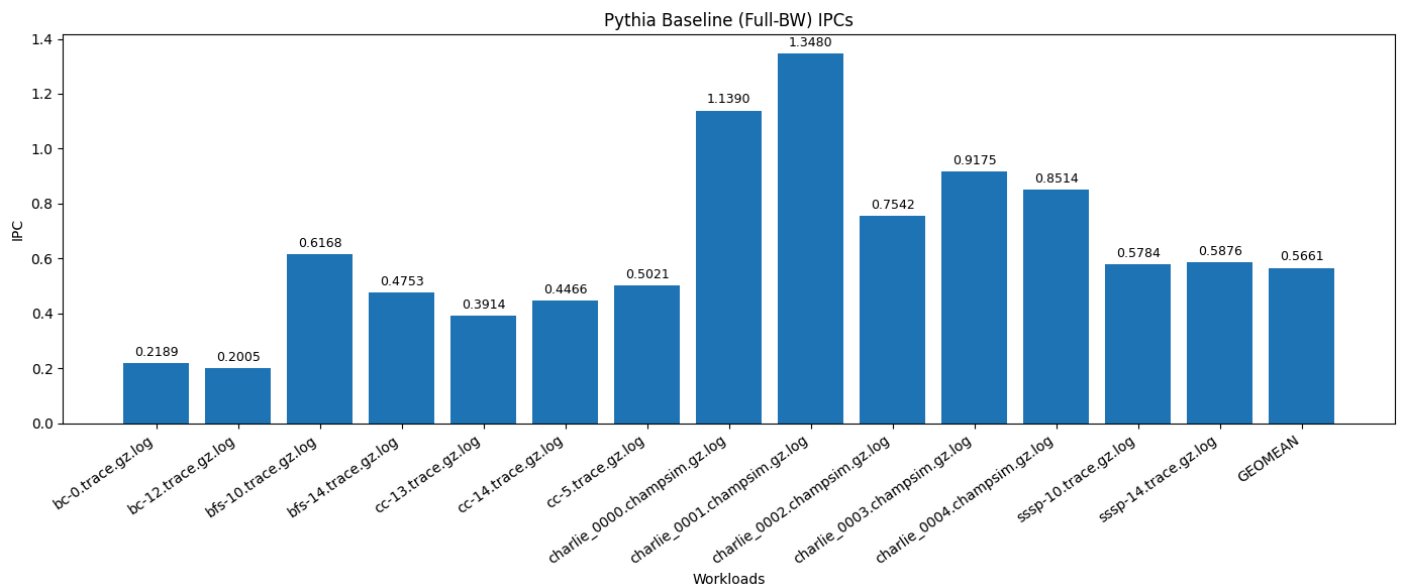
Trace	Speedup
bc-0.trace.gz.log	1.1131
bc-12.trace.gz.log	1.0942
bfs-10.trace.gz.log	1.3344
bfs-14.trace.gz.log	1.5035
cc-13.trace.gz.log	1.3306
cc-14.trace.gz.log	1.3576
cc-5.trace.gz.log	1.3763
charlie_0000.champsim.gz.log	1.0071
charlie_0001.champsim.gz.log	1.0124
charlie_0002.champsim.gz.log	1.0038
charlie_0003.champsim.gz.log	0.9955
charlie_0004.champsim.gz.log	1.0029
sssp-10.trace.gz.log	1.3138
sssp-14.trace.gz.log	1.3225
GEOMEAN	1.1852



3.6. Bonus (+20%) Task: Comparison Against a State-Of-The-Art Prefetcher

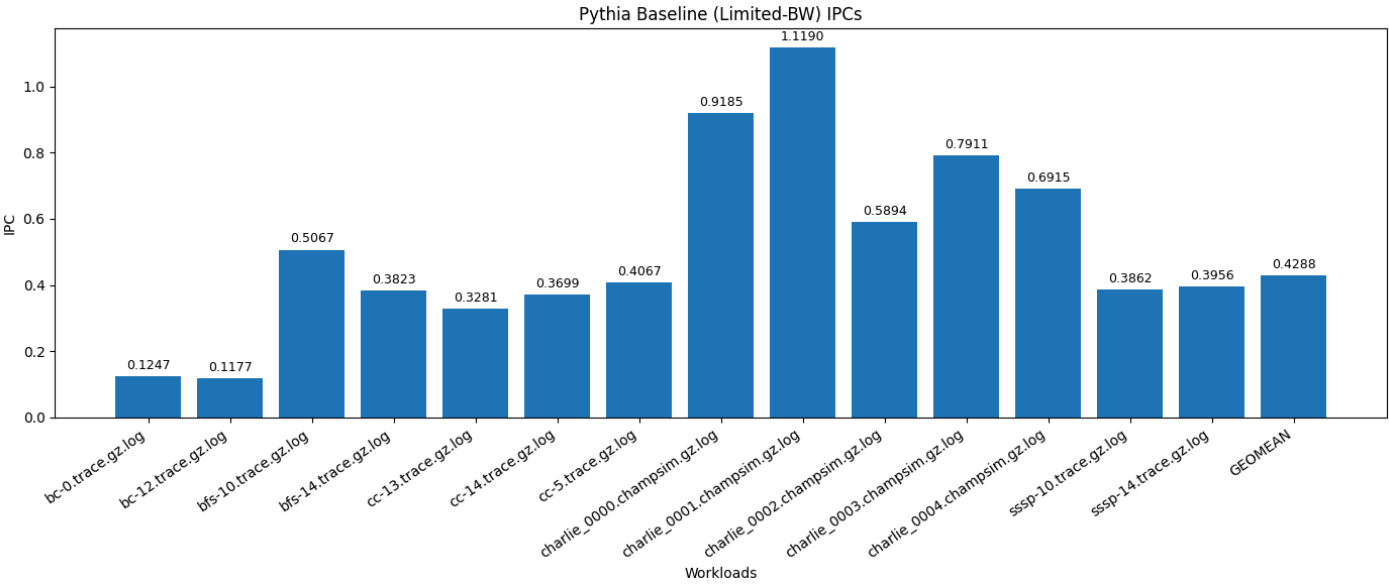
Pythia Baseline (Full-BW) IPCs

Trace	IPC
bc-0.trace.gz.log	0.2189
bc-12.trace.gz.log	0.2005
bfs-10.trace.gz.log	0.6168
bfs-14.trace.gz.log	0.4753
cc-13.trace.gz.log	0.3914
cc-14.trace.gz.log	0.4466
cc-5.trace.gz.log	0.5021
charlie_0000.champsim.gz.log	1.1390
charlie_0001.champsim.gz.log	1.3480
charlie_0002.champsim.gz.log	0.7542
charlie_0003.champsim.gz.log	0.9175
charlie_0004.champsim.gz.log	0.8514
sssp-10.trace.gz.log	0.5784
sssp-14.trace.gz.log	0.5876
GEOMEAN	0.5661



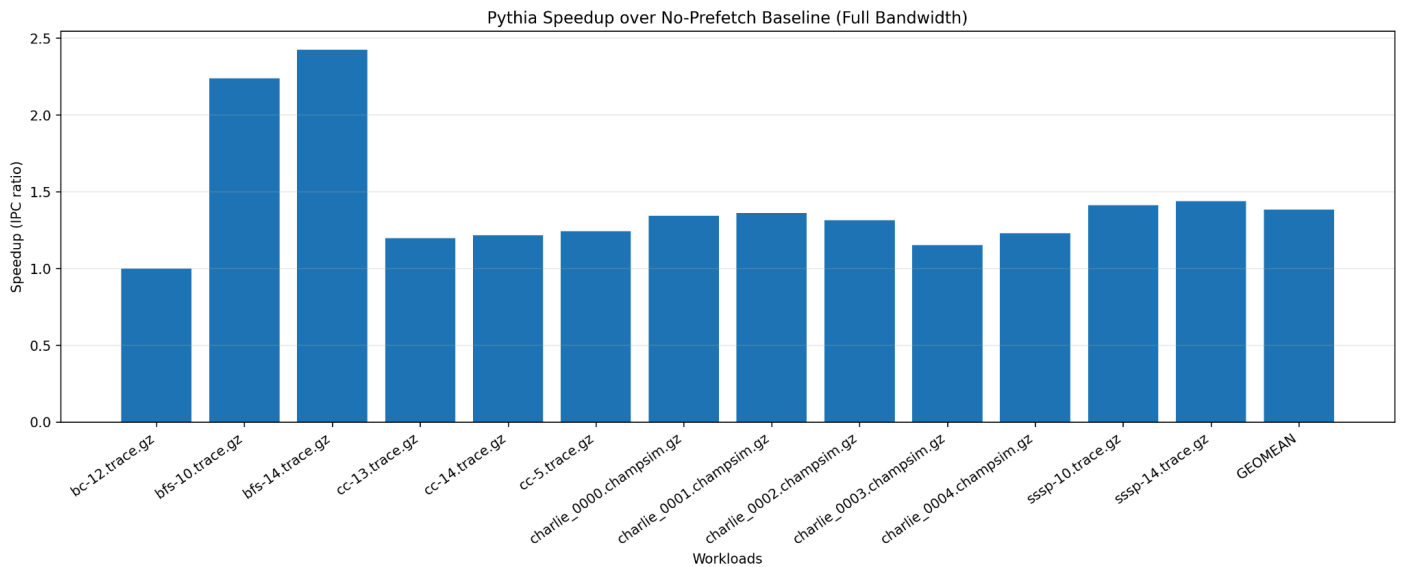
Pythia Baseline (Limited-BW) IPCs

Trace	IPC
bc-0.trace.gz.log	0.1247
bc-12.trace.gz.log	0.1177
bfs-10.trace.gz.log	0.5067
bfs-14.trace.gz.log	0.3823
cc-13.trace.gz.log	0.3281
cc-14.trace.gz.log	0.3699
cc-5.trace.gz.log	0.4067
charlie_0000.champsim.gz.log	0.9185
charlie_0001.champsim.gz.log	1.1190
charlie_0002.champsim.gz.log	0.5894
charlie_0003.champsim.gz.log	0.7911
charlie_0004.champsim.gz.log	0.6915
sssp-10.trace.gz.log	0.3862
sssp-14.trace.gz.log	0.3956
GEOMEAN	0.4288



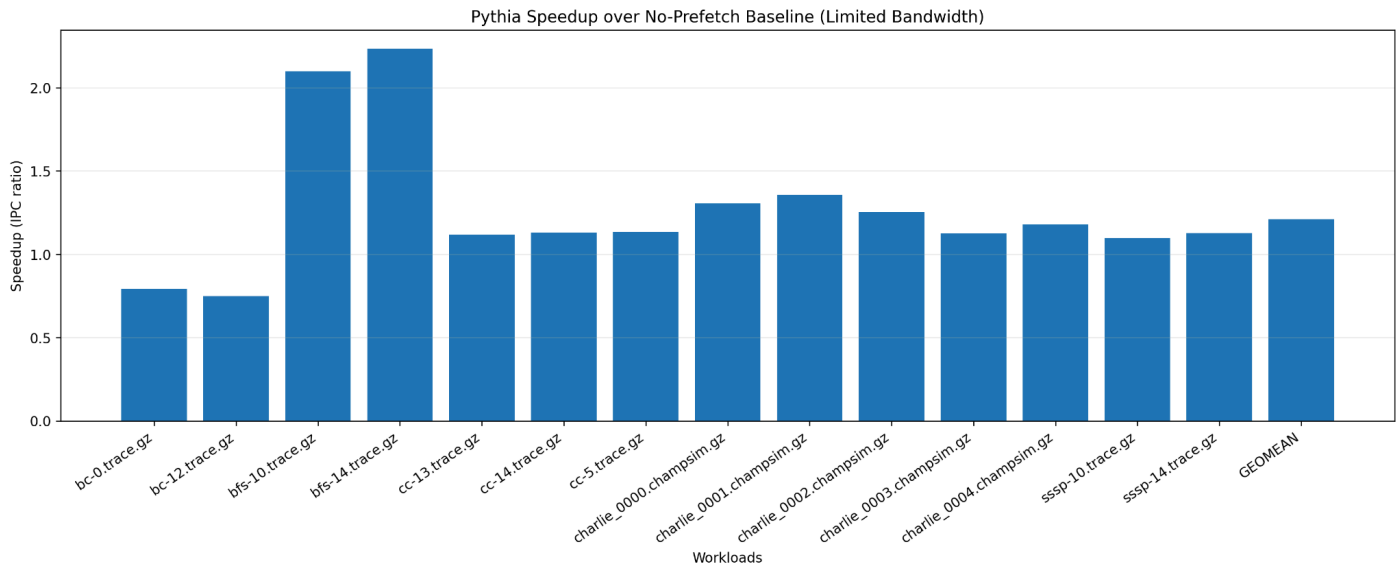
Pythia Speedup over No-Prefetch Baseline (Full Bandwidth)

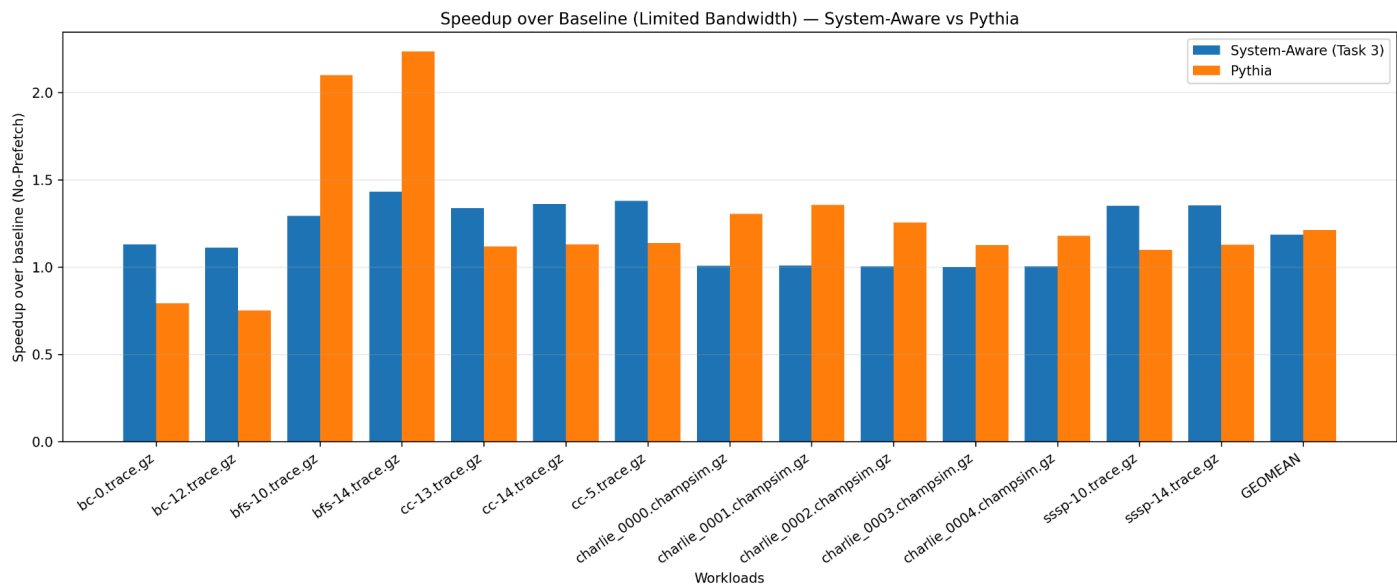
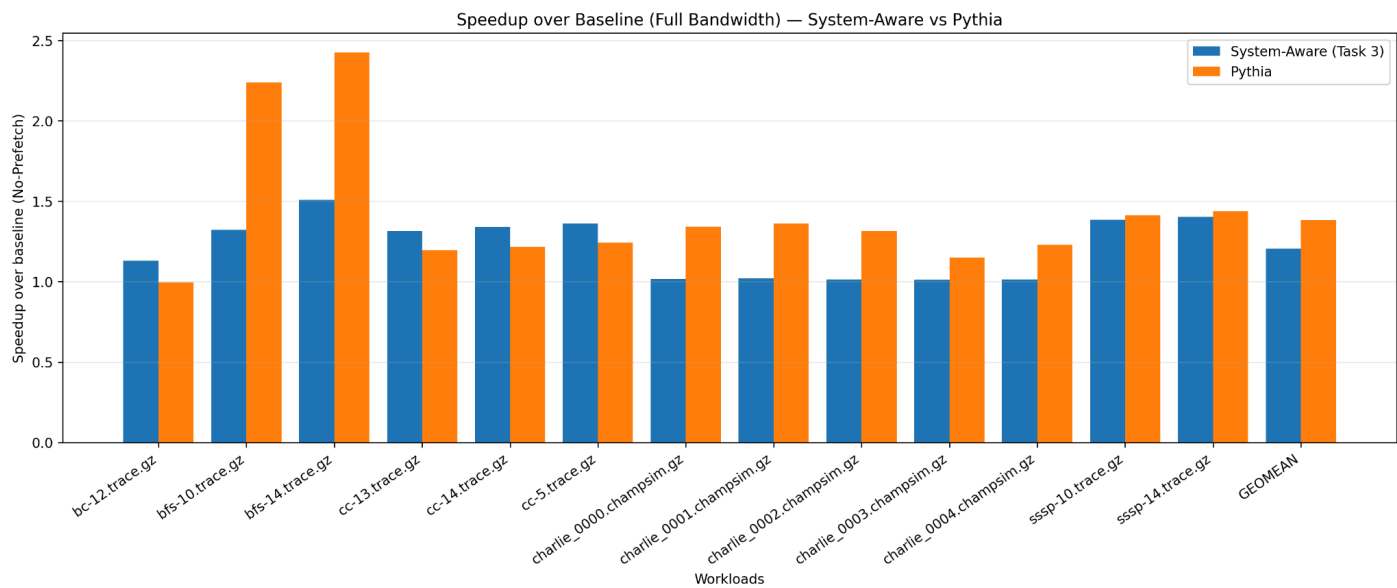
Trace	Baseline IPC	Pythia IPC	Speedup
bc-12.trace.gz	0.2011	0.2005	0.9970
bfs-10.trace.gz	0.2755	0.6168	2.2388
bfs-14.trace.gz	0.1961	0.4753	2.4238
cc-13.trace.gz	0.3273	0.3914	1.1958
cc-14.trace.gz	0.3672	0.4466	1.2162
cc-5.trace.gz	0.4039	0.5021	1.2431
charlie_0000.champsim.gz	0.8490	1.1390	1.3416
charlie_0001.champsim.gz	0.9901	1.3480	1.3615
charlie_0002.champsim.gz	0.5743	0.7542	1.3133
charlie_0003.champsim.gz	0.7972	0.9175	1.1509
charlie_0004.champsim.gz	0.6932	0.8514	1.2282
sssp-10.trace.gz	0.4097	0.5784	1.4118
sssp-14.trace.gz	0.4087	0.5876	1.4377
GEOMEAN			1.3824



Pythia Speedup over No-Prefetch Baseline (Limited Bandwidth)

Trace	Baseline IPC	Pythia IPC	Speedup
bc-0.trace.gz	0.1574	0.1247	0.7922
bc-12.trace.gz	0.1571	0.1177	0.7492
bfs-10.trace.gz	0.2413	0.5067	2.0999
bfs-14.trace.gz	0.1712	0.3823	2.2331
cc-13.trace.gz	0.2937	0.3281	1.1171
cc-14.trace.gz	0.3275	0.3699	1.1295
cc-5.trace.gz	0.3582	0.4067	1.1354
charlie_0000.champsim.gz	0.7041	0.9185	1.3045
charlie_0001.champsim.gz	0.8252	1.1190	1.3560
charlie_0002.champsim.gz	0.4697	0.5894	1.2548
charlie_0003.champsim.gz	0.7037	0.7911	1.1242
charlie_0004.champsim.gz	0.5868	0.6915	1.1784
sssp-10.trace.gz	0.3518	0.3862	1.0978
sssp-14.trace.gz	0.3510	0.3956	1.1271
GEOMEAN			1.2105





Prefetcher comparison notes:

- No prefetcher baseline (full BW): IPCs hover between ~0.20 (bfs/cc) and ~0.99 (charlie). This is the reference for all speedups in Tasks 1/2/3 and the bonus.
- GHB-stride (full BW): Accelerates every workload, with per-trace speedups ranging from ~1.01 (charlie) to ~1.41 (sssp). The geomean is comfortably above 1, showing the stride pattern detection pays off with plentiful bandwidth.
- GHB-stride (limited BW): IPCs drop across the board, and the limited-bandwidth speedups shrink compared to the full-bandwidth case - some speedups even fall below 1 for specific workloads - because inaccurate prefetches steal scarce DRAM cycles. That's exactly why Task 3 introduced the system-aware variant.
- GHB-stride system-aware: By tracking pf-issued/pf-useful and DRAM bandwidth, it throttles the degree when accuracy falls or bandwidth is saturated, keeping the speedups closer to the full-bandwidth numbers even under constrained bandwidth. It still outperforms the no-prefetcher baseline, and the behavior across workloads is more stable than the vanilla GHB-stride limited-BW results.
- Pythia (bonus): With reinforcement-learning steering prefetch decisions, Pythia posts the highest IPCs in both bandwidth regimes (e.g., cc and charlie traces exceed 1.0 on full BW). Speedups over the no-prefetcher baseline are the largest of any design we've tried, so Pythia is the strongest performer, especially when enough bandwidth exists to absorb its speculative requests.